

General Waveform Decomposition for Neural Network Emulation

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Abstract

We present a general framework for neural network emulation of gravitational waveforms that can accommodate arbitrary physical complexity: precession, higher modes, and eccentricity. We clarify which aspects of the standard decomposition are coordinate choices versus restrictive physical assumptions, and propose a neural network architecture that learns the optimal frame decomposition end-to-end without requiring external Euler angle extraction.

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1 Introduction

Existing neural network emulators for gravitational waveforms make simplifying assumptions that limit their generality:

- **mlgw_NN** [1]: Assumes aligned spins (non-precessing) and circular orbits
- **SEOBNN** [2]: Assumes single-spin systems ($\vec{\chi}_2 = 0$), conjugate mode symmetry in the coprecessing frame, and circular orbits
- **SEOBNRE_AI** [3]: Includes eccentricity but only models the dominant (2,2) mode with aligned spins

This document presents a framework that can accommodate any level of complexity, and addresses practical concerns about deploying such a framework on arbitrary waveform models.

The approach of decomposing waveforms into coprecessing-frame modes plus frame dynamics has a long history in the numerical relativity surrogate community [4, 5, 6, 7], and the minimal rotation frame was established by [7, 8]. This document aims to synthesise these ideas into a unified framework for neural network emulation.

2 Conventions

We adopt the following conventions throughout:

Complex strain. The complex gravitational wave strain is defined as

$$h \equiv h_+ - ih_\times \quad (1)$$

Under a polarisation rotation by angle ψ , this transforms as $h \rightarrow e^{-2i\psi}h$.

Euler angles. We use the z - y - z (ZYZ) active rotation convention. The rotation $R(\alpha, \beta, \gamma)$ is interpreted as: first rotate by γ about z , then by β about the new y , then by α about the final z . This convention is standard in the gravitational wave literature [7, 8]. We treat $R(\alpha, \beta, \gamma)$ as an *active* rotation acting on the physical field; equivalently, it corresponds to a passive change of basis by R^{-1} . All formulas below use the active convention.

Wigner D-matrices. For active rotations, the spherical harmonic modes transform as

$$h_{\ell m}^{\text{rotated}} = \sum_{m'=-\ell}^{+\ell} D_{mm'}^\ell(\alpha, \beta, \gamma) h_{\ell m'} \quad (2)$$

where $D_{mm'}^\ell(\alpha, \beta, \gamma) = e^{-im\alpha} d_{mm'}^\ell(\beta) e^{-im'\gamma}$ and $d_{mm'}^\ell(\beta)$ are the reduced Wigner matrices. We use the standard GW convention for ${}^{-2}Y_{\ell m}$ as in [7]; with this choice the modes mix under rotations according to (2).

Inertial frame. The inertial-frame modes $h_{\ell m}$ are defined with respect to an inertial triad whose $\hat{z}$ axis is aligned with $\hat{J}(t_{\text{ref}})$, the total angular momentum at a reference time. The inclination angle ι and reference azimuth φ_0 are the spherical angles of the line-of-sight vector in this triad. The choice $\iota = 0$ corresponds to “face-on” (viewing along $\hat{J}$).

Time and mass scaling. Unless otherwise stated, time t is in units of the total mass M (i.e., dimensionless time t/M in geometric units $G = c = 1$). Physical time in seconds is recovered via $t_{\text{phys}} = (M/M_\odot) \times 4.926 \times 10^{-6} \text{ s} \times t$.

Coalescence time and phase. The coalescence time t_c and coalescence phase ϕ_c are extrinsic parameters that shift the waveform in time and rotate the complex modes: $h_{\ell m}(t) \rightarrow h_{\ell m}(t - t_c)e^{-im\phi_c}$ (sign consistent with (2) and the LAL/Boyle convention [7]).

Phase parameter degeneracy. The reference azimuth φ_0 and coalescence phase ϕ_c are convention-dependent and partially degenerate: a shift in one can be compensated by a shift in the other. In practice, we fix one by a reference-phase alignment choice (e.g., $\phi_{22}(t_{\text{ref}}) = 0$).

3 General Decomposition

3.1 Spherical Harmonic Expansion

The gravitational wave strain observed at angles (ι, φ_0) from the source is expanded in spin-weighted spherical harmonics:

$$h(t; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell=2}^{\ell_{\text{max}}} \sum_{m=-\ell}^{+\ell} h_{\ell m}(t; \boldsymbol{\lambda}) {}^{-2}Y_{\ell m}(\iota, \varphi_0) \quad (3)$$

where ${}^{-2}Y_{\ell m}$ are the spin -2 weighted spherical harmonics, and $\boldsymbol{\lambda}$ denotes the *intrinsic* parameter space.

3.2 Intrinsic vs Extrinsic Parameters

A crucial feature of the decomposition (3) is that it separates parameters into two classes:

Intrinsic parameters $\boldsymbol{\lambda}$: Properties of the source that affect the mode functions $h_{\ell m}(t)$:

$$\boldsymbol{\lambda} = \{q, \chi_{1x}, \chi_{1y}, \chi_{1z}, \chi_{2x}, \chi_{2y}, \chi_{2z}, e_0, \dots\} \quad (4)$$

where $q = m_1/m_2$ is the mass ratio, $\vec{\chi}_1, \vec{\chi}_2$ are the dimensionless spin vectors, and e_0 is the orbital eccentricity.

Extrinsic parameters: Properties that affect only how the modes combine or scale:

- **Inclination ι :** Enters only through ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$
- **Reference phase φ_0 :** Enters only through ${}^{-2}Y_{\ell m}(\iota, \varphi_0)$
- **Luminosity distance d_L :** Scales amplitude as $h \propto 1/d_L$
- **Polarisation angle ψ :** Rotates $h_+, h_\times$ at the detector

Implication for emulation: The neural network only needs to learn the intrinsic modes $h_{\ell m}(t; \boldsymbol{\lambda})$. Extrinsic parameters are handled analytically at inference time via the known spherical harmonic functions. One trained model works for all viewing angles.

3.3 Coprecessing Frame Decomposition

For precessing systems, we decompose each inertial-frame mode into coprecessing-frame modes via a time-dependent rotation:

$$h_{\ell m}(t; \boldsymbol{\lambda}) = \sum_{m'=-\ell}^{+\ell} D_{mm'}^\ell(\alpha(t), \beta(t), \gamma(t)) h_{\ell m'}^{\text{cp}}(t; \boldsymbol{\lambda}) \quad (5)$$

where $D_{mm'}^\ell(\alpha, \beta, \gamma) = e^{-im\alpha} d_{mm'}^\ell(\beta) e^{-im'\gamma}$ are the Wigner D-matrices (see §2). The Euler angles $\{\alpha(t), \beta(t), \gamma(t)\}$ define a time-dependent rotation from the coprecessing frame to the inertial frame.

The coprecessing frame is typically chosen to track the instantaneous orbital angular momentum $\hat{L}(t)$, though alternative definitions exist (see §6.2). For non-precessing systems, the rotation is trivial ($D_{mm'}^\ell \rightarrow \delta_{mm'}$) and the coprecessing modes equal the inertial modes.

3.4 Amplitude–Phase Decomposition

Each coprecessing mode is decomposed into amplitude and phase:

$$h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}) = A_{\ell m}(t; \boldsymbol{\lambda}) e^{i\phi_{\ell m}(t; \boldsymbol{\lambda})} \quad (6)$$

This decomposition is advantageous because $A_{\ell m}(t)$ and $\phi_{\ell m}(t)$ are smooth functions that are easier for neural networks to learn than the oscillatory complex waveform.

Practical considerations. Several subtleties arise in practice:

- **Phase unwrapping:** The phase $\phi_{\ell m}(t)$ must be unwrapped to avoid 2π discontinuities. This requires careful handling, especially near merger.
- **Log amplitude:** Learning $\log A_{\ell m}(t)$ rather than $A_{\ell m}(t)$ can improve conditioning, particularly given the ~ 5 orders of magnitude dynamic range.
- **Amplitude zeros:** When $A_{\ell m}(t) \rightarrow 0$, the phase becomes ill-defined. Modes with amplitude zeros require special treatment.
- **Instantaneous frequency:** For some applications, learning $\dot{\phi}_{\ell m}(t)$ and integrating may be more stable than learning $\phi_{\ell m}(t)$ directly.

4 What is a Coordinate Choice vs a Restrictive Assumption

4.1 Coordinate Choices (Not Restrictive)

The following are *coordinate transformations* that can be applied to any waveform without loss of generality. They are always *mathematically* valid, though some choices are more *numerically* stable for machine learning than others.

1. **Spherical harmonic expansion** (3): Always valid for any radiation field. The expansion converges for smooth functions on the sphere.
2. **Coprecessing frame rotation** (5): This is a time-dependent rotation of coordinates. While mathematically always valid and invertible, the *choice* of frame convention affects numerical stability:
 - Some frames (e.g., the instantaneous $\hat{L}(t)$ frame) can exhibit rapid variations near merger or during transitional precession.
 - The minimal rotation frame [7] imposes $\dot{\gamma} + \dot{\alpha} \cos \beta = 0$, reducing frame oscillations.
 - Different choices give different Euler angles but are all physically equivalent.
3. **Amplitude–phase decomposition** (6): Any complex function can be written in polar form. However, this representation is ill-conditioned when amplitudes approach zero (phase undefined) or when phase wrapping occurs.

4.2 Restrictive Assumptions

The following are *physical approximations* that limit the class of waveforms that can be accurately represented:

1. **Conjugate symmetry in the inertial frame** (aligned-spin assumption):

$$h_{\ell,-m}(t; \boldsymbol{\lambda}) = (-1)^\ell h_{\ell m}^*(t; \boldsymbol{\lambda}) \quad (7)$$

This holds only for non-precessing systems with aligned spins.

2. **Conjugate symmetry in the coprecessing frame:**

$$h_{\ell,-m}^{\text{cp}}(t; \boldsymbol{\lambda}) = (-1)^\ell (h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}))^* \quad (8)$$

This approximation assumes the coprecessing-frame waveform looks like an aligned-spin waveform. It works well for mild precession but breaks down for strongly precessing systems.

3. **Single-spin assumption** ($\vec{\chi}_2 = 0$): Reduces the parameter space but excludes double-spin systems.
4. **Circular orbit assumption** ($e_0 = 0$): Excludes eccentric binaries.

4.3 Summary Table

Component	Type	Always Valid?
Spherical harmonic expansion	Coordinate choice	Yes
Coprecessing frame rotation	Coordinate choice	Yes
Amplitude/phase split	Coordinate choice	Yes
Conjugate symmetry (inertial)	Approximation	No (aligned spin only)
Conjugate symmetry (coprecessing)	Approximation	No (mild precession only)
Single spin	Approximation	No
Circular orbits	Approximation	No

Table 1: Classification of decomposition components.

5 Where Different Physics Enters

5.1 Precession

Precession is handled by the coprecessing frame transformation. The Euler angles $\alpha(t), \beta(t), \gamma(t)$ encode the precession dynamics—the time-dependent orientation of the orbital plane.

Key point: The coprecessing frame transformation *handles* precession but doesn't *add* it. If the underlying modes don't contain precession physics, the transformation is trivial ($D_{mm'}^\ell \rightarrow \delta_{mm'}$).

5.2 Eccentricity

Eccentricity affects the *intrinsic* time evolution of the modes. It enters through:

1. Mode amplitudes $A_{\ell m}(t)$: Eccentric orbits produce amplitude modulations at the orbital and radial frequencies, and excite additional harmonic content (sidebands).

2. Mode phases $\phi_{\ell m}(t)$: More complex orbital phase evolution with periastron advance.
3. Mode structure: Eccentric waveforms have power in many more (ℓ, m) modes than circular waveforms at the same orbital frequency.

Eccentricity and precession. When both eccentricity and precession are present, the Euler angles $\alpha(t), \beta(t), \gamma(t)$ develop additional modulations. However, for non-precessing eccentric systems (aligned spins), the orbital plane is fixed and the Euler angles are constant—eccentricity alone does not induce frame rotation.

Conjugate symmetry and eccentricity. The conjugate symmetry relation (7) depends on reflection symmetry about the orbital plane. For non-precessing systems with aligned spins, this symmetry is preserved even with eccentricity. Eccentricity alone does not break conjugate symmetry; precession does.

Key point: Eccentricity cannot be added by coordinate transformations. It must be present in the underlying waveform model being emulated.

5.3 Implications for IMRPhenomX Family

- **IMRPhenomXHM:** Non-precessing, higher modes, circular. Can serve as coprecessing modes.
- **IMRPhenomXPHM:** Uses XHM as coprecessing modes + Euler angle rotation. Precessing, higher modes, circular.
- **Neither has eccentricity:** To emulate eccentric waveforms, one must start from an eccentric model (e.g., SEOBNRE, TEOBResumS-GIOTTO).

6 The Euler Angle Extraction Problem

6.1 The Practical Challenge

For emulation, we assume the input is a “black box” waveform code that outputs only inertial-frame quantities:

$$\text{Code : } \lambda \mapsto \{h_+(t), h_\times(t)\} \quad \text{or} \quad \lambda \mapsto \{h_{\ell m}(t)\} \quad (9)$$

The coprecessing modes and Euler angles are *not* provided. Some codes (e.g., SEOB-NRv4PHM in LALSuite) do expose these internally, but this cannot be assumed in general.

6.2 Non-Uniqueness of the Coprecessing Frame

There is no single “correct” definition of the coprecessing frame. Valid choices include:

- **L -frame:** Track instantaneous orbital angular momentum $\hat{L}(t)$
- **J -frame:** Align with total angular momentum $\hat{J}$ at reference time
- **Quadrupole-preferred frame:** Minimise power in non- $(2, \pm 2)$ modes
- **Minimal rotation frame:** Impose $\dot{\gamma} = -\dot{\alpha} \cos \beta$

Different conventions give different Euler angles and coprecessing modes, though all are valid coordinate choices.

6.3 Classical Extraction Methods

If one chooses to extract Euler angles from inertial-frame waveforms, established methods exist:

1. **Dominant eigenvector method:** Find the frame maximising power in $(2, \pm 2)$
2. **Phase gradient method:** Extract from the $(2, 2)$ mode phase evolution
3. **Optimisation:** Minimise some cost function over rotation parameters

These methods work but introduce preprocessing complexity and convention dependence.

7 Neural Network Architecture Options

We consider three architectural approaches for learning precessing waveforms.

7.1 Option 1: Extract Then Learn

1. Use classical algorithms to extract Euler angles from training waveforms
2. Train separate neural networks for coprecessing modes and Euler angles
3. At inference, predict both and combine via Wigner rotation

Pros: Physics-informed, interpretable, follows literature

Cons: Extraction is non-unique, adds preprocessing step, convention-dependent

7.2 Option 2: Learn Inertial Modes Directly

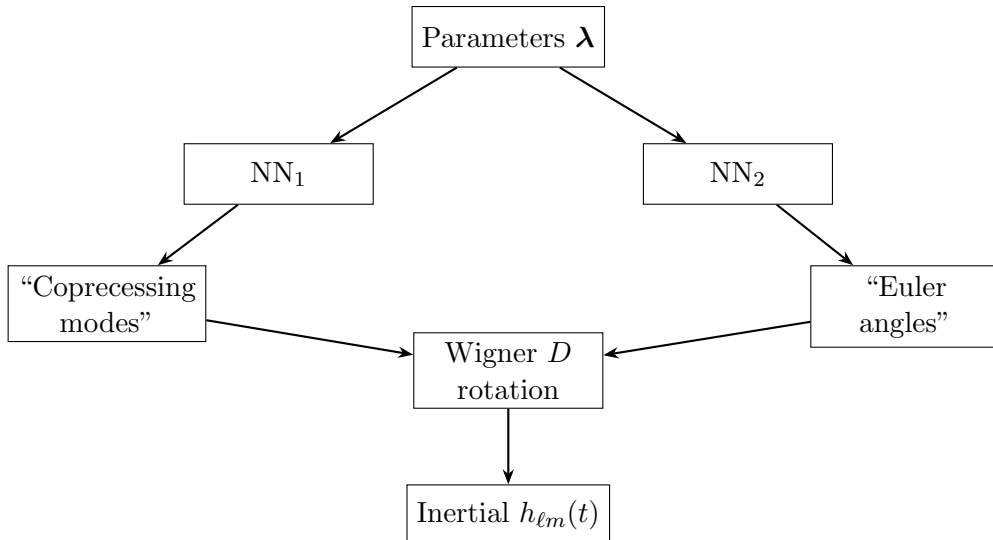
1. Skip decomposition entirely
2. Train neural network: $\lambda \mapsto h_{\ell m}(t)$ in inertial frame

Pros: Simple pipeline, no extraction needed

Cons: Inertial modes have fast precession oscillations, much harder to learn, requires larger networks

7.3 Option 3: Learn Frame Decomposition End-to-End (Recommended)

Build the Wigner rotation into the network architecture:



The loss function is computed on the inertial-frame output:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \mathcal{L}_{\text{reg}} \quad (10)$$

where the data term is

$$\mathcal{L}_{\text{data}} = \sum_{\ell m} \left\| h_{\ell m}^{\text{pred}}(t) - h_{\ell m}^{\text{true}}(t) \right\|^2 \quad (11)$$

The gauge problem. A critical issue with Option 3 is **gauge non-identifiability**: the decomposition into Euler angles and coprecessing modes is not unique. Without constraints, the network could learn a pathological decomposition—for example, setting $\alpha = \beta = \gamma = 0$ and absorbing all oscillatory behaviour into the “coprecessing” modes. This defeats the purpose of the architecture, which aims to make the coprecessing modes slowly varying.

Regularization. To encourage meaningful decompositions, we add a regularization term:

$$\mathcal{L}_{\text{reg}} = \lambda_1 \mathcal{L}_{\text{mode}} + \lambda_2 \mathcal{L}_{\text{angle}} + \lambda_3 \mathcal{L}_{\text{minrot}} + \lambda_4 \mathcal{L}_{\text{anchor}} \quad (12)$$

where:

$$\mathcal{L}_{\text{mode}} = \sum_{\ell m} \left(\left\| \ddot{A}_{\ell m} \right\|^2 + \left\| \ddot{\omega}_{\ell m} \right\|^2 \right) \quad (\text{smooth coprecessing modes}) \quad (13)$$

$$\mathcal{L}_{\text{angle}} = \left\| \dot{\alpha} \right\|^2 + \left\| \dot{\beta} \right\|^2 + \left\| \dot{\gamma} \right\|^2 \quad (\text{slow angle evolution}) \quad (14)$$

$$\mathcal{L}_{\text{minrot}} = \left\| \dot{\gamma} + \dot{\alpha} \cos \beta \right\|^2 \quad (\text{minimal rotation [7]}) \quad (15)$$

$$\mathcal{L}_{\text{anchor}} = \alpha(t_{\text{ref}})^2 + \gamma(t_{\text{ref}})^2 \quad (\text{frame anchoring}) \quad (16)$$

Here $\omega_{\ell m} = \dot{\phi}_{\ell m}$ is the instantaneous frequency of each coprecessing mode.

The mode regularization penalises rapid oscillations in coprecessing amplitudes and *instantaneous frequencies* (not raw phases). This is essential: the phase $\phi_{\ell m}(t)$ naturally has large curvature during inspiral (the chirp), which is physical. Regularizing ϕ directly would fight real dynamics. Instead, regularizing $\ddot{\omega} = \ddot{\phi}$ encourages smooth frequency evolution without penalising the secular chirp.

The angle regularization penalises angular *velocities* $\dot{\alpha}, \dot{\beta}, \dot{\gamma}$, ensuring the Euler angles vary on the slow precession timescale rather than stealing the fast orbital frequency. The minimal-rotation term encourages the standard frame convention. The anchor term fixes the global rotational freedom at the reference time, ensuring a unique decomposition.

Derivatives: When using basis-coefficient outputs, derivatives in $\mathcal{L}_{\text{reg}}$ should be computed analytically from the basis functions $B_k(t)$ rather than finite differences, for numerical stability.

Angle representation. Directly predicting angles $\alpha, \gamma \in [0, 2\pi)$ is numerically fragile due to the discontinuity at 2π . Instead, the network should output:

$$\text{For } \alpha, \gamma : \quad (\sin \alpha, \cos \alpha), \quad (\sin \gamma, \cos \gamma) \quad (17)$$

$$\text{For } \beta : \quad \cos \beta \in [-1, 1] \quad (\text{or } \beta \text{ directly since } \beta \in [0, \pi]) \quad (18)$$

Alternatively, quaternion representations avoid gimbal lock and singularities entirely.

Time-series parameterization. The network does not output raw time series $A_{\ell m}(t), \phi_{\ell m}(t), \alpha(t)$, etc. directly. Instead, outputs are *coefficients* in a reduced basis:

$$A_{\ell m}(t) = \sum_{k=1}^K c_{\ell m, k}^{(A)} B_k(t) \quad (19)$$

where $B_k(t)$ are basis functions (e.g., from PCA/SVD of training data, empirical interpolation nodes, or spline control points). The network predicts the coefficients $c_{\ell m, k}^{(A)}$. This reduces the output dimensionality from $\mathcal{O}(N_t)$ to $\mathcal{O}(K)$ per function, where typically $K \ll N_t$.

Initialization and training stability. The multiplicative structure of Option 3 (Wigner D-matrix combining outputs of two networks) makes training from random initialisation challenging. If the angle-prediction branch outputs random noise at epoch 0, gradients flowing to the mode-prediction branch become chaotic, often causing the network to collapse to a trivial decomposition ($\alpha = \beta = \gamma = 0$).

Recommended initialization strategy:

- Initialize the angle-prediction branch to output the identity rotation: $(\sin \alpha, \cos \alpha) \rightarrow (0, 1)$, etc.
- This encourages the network to first learn an approximate waveform in the “coprocessing” branch, then gradually “turn on” the rotation to reduce loss.
- Alternatively, warm-start with a Newtonian chirp approximation for the angles.

Data preprocessing. Training data must be consistently aligned:

- **Time alignment:** Align waveforms to a common reference (e.g., peak of $|h_{22}|$, or merger time).
- **Phase alignment:** Fix the orbital phase at reference time (e.g., $\phi_{22}(t_{\text{ref}}) = 0$).
- **Frame anchoring:** The training targets should be consistent with the anchor constraint $\alpha(t_{\text{ref}}) = \gamma(t_{\text{ref}}) = 0$.

Basis considerations for eccentricity. SVD/PCA bases rely on features appearing at consistent times. Eccentricity introduces frequency shifts that can de-phase waveforms significantly across the training set. For eccentric waveforms, the basis $B_k(t)$ may need to be derived from phase-aligned (rather than time-aligned) waveforms, or a larger basis dimension K may be required.

Key features:

- No external Euler angle extraction needed
- Physics structure (rotation equivariance) is built into architecture
- Network learns whatever internal decomposition works best
- The learned “Euler angles” are latent variables—they need not match any standard convention
- Works with any code that outputs $h_{\ell m}(t)$ in the inertial frame
- Regularization encourages physically meaningful decomposition

Pros: General, no preprocessing, physics-informed architecture

Cons: More complex architecture, requires careful regularization, learned angles may not be interpretable

Criterion	Option 1	Option 2	Option 3
External extraction needed	Yes	No	No
Convention dependence	Yes	No	Implicit [†]
Physics structure exploited	Yes	No	Yes
Interpretable intermediates	Yes	N/A	No
Network complexity	Medium	High	Medium
Training difficulty	Medium	Hard	Medium

Table 2: Comparison of architectural options. [†]Option 3 still depends on conventions through the Wigner-D definition, basis representation, and regularization choices, but these are implicit rather than requiring explicit frame extraction.

7.4 Comparison

8 Architecture Requirements

8.1 Without Conjugate Symmetry

For a fully general emulator covering $\ell \leq \ell_{\max}$ without assuming conjugate symmetry:

Component	Count	Notes
Coprocessing amplitudes $A_{\ell m}(t)$	$\sum_{\ell=2}^{\ell_{\max}} (2\ell + 1)$	All $m \in [-\ell, +\ell]$
Coprocessing phases $\phi_{\ell m}(t)$	$\sum_{\ell=2}^{\ell_{\max}} (2\ell + 1)$	All $m \in [-\ell, +\ell]$
Euler angles	3	$\alpha(t), \beta(t), \gamma(t)$

Table 3: Neural network outputs required for general emulation.

For $\ell_{\max} = 4$ (modes up to $(4, \pm 4)$): $5 + 7 + 9 = 21$ complex modes $\times 2$ (amp/phase) + 3 Euler angles = 45 output functions.

8.2 With Conjugate Symmetry (if applicable)

If conjugate symmetry can be assumed (aligned spins or mild precession), only $m \geq 0$ modes need to be modelled, roughly halving the outputs.

9 Summary

The complete general decomposition is:

$$h(t; \boldsymbol{\lambda}; \iota, \varphi_0) = \sum_{\ell, m} h_{\ell m}(t; \boldsymbol{\lambda})^{-2} Y_{\ell m}(\iota, \varphi_0) \quad (20)$$

$$h_{\ell m}(t; \boldsymbol{\lambda}) = \sum_{m'=-\ell}^{+\ell} D_{mm'}^{\ell}(\alpha(t), \beta(t), \gamma(t)) h_{\ell m'}^{\text{cp}}(t; \boldsymbol{\lambda}) \quad (21)$$

$$h_{\ell m}^{\text{cp}}(t; \boldsymbol{\lambda}) = A_{\ell m}(t; \boldsymbol{\lambda}) e^{i\phi_{\ell m}(t; \boldsymbol{\lambda})} \quad (22)$$

Key points:

1. Inclination and reference phase are extrinsic—handled analytically via spherical harmonics.
2. The coprocessing frame decomposition is a coordinate choice, not a restrictive assumption—though the choice of frame convention affects numerical stability.

3. Conjugate symmetry (in either frame) IS a restrictive assumption.
4. Eccentricity must be in the underlying model—it cannot be added by coordinate transformations. Eccentricity alone does not break conjugate symmetry; precession does.
5. For practical emulation, we recommend learning the frame decomposition end-to-end (Option 3), with appropriate regularization to address gauge non-identifiability.
6. Euler angles should be represented as $(\sin \alpha, \cos \alpha)$ pairs to avoid 2π discontinuities.
7. Network outputs should be basis coefficients, not raw time series.

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